

Airpay Academy 2.0 — Upgrade Summary

What Changed from Current Production to v1.0.0-rc1

Date: April 2026 **From:** Moodle 4.1 + epsilon theme + BizLMS (as-is from production) **To:** Moodle 4.5.10 + airpayux theme + BizLMS (patched) + 4 custom plugins

Executive Summary

Academy 2.0 is a complete visual and functional overhaul of the Airpay Academy LMS. It replaces the default epsilon theme with a custom-built airpayux theme, adds 4 new plugins, fixes 30+ BizLMS bugs, and introduces role-based dashboards, dark mode, mobile responsiveness, and tenant-scoped administration.

No database schema changes required. All upgrades are file-level (theme, plugins, templates). The existing production database works as-is.

What's New

1. Custom Theme: airpayux

Feature	Before (epsilon)	After (airpayux)
Login page	Basic Moodle form	Split-screen with logo, stats, branding
Dashboard	Generic block layout	Role-based (4 tiers: Admin, L&D, Manager, Learner)
Navbar	Standard Moodle nav	Pill navigation + Quick Access + Dark mode toggle
Footer	Standard Moodle footer	Compact branded footer (logo, links, Made in India)
Dark mode	None	Full dark mode on all pages with localStorage persistence
Mobile	Basic responsive	Optimized for 768/590/480px with BizLMS-specific rules
Branding	Generic	Dynamic tenant logos from costcenter_logo field
Homepage	Site info page	Enterprise landing page with hero, stats, CTAs

2. Role-Based Dashboards

Role	What They See
Siteadmin	Global KPIs + Charts + Quick Nav + System Health + User Analytics
L&D Admin	Tenant-scoped KPIs + Charts + Quick Nav (NO System Health)
Manager	Team KPIs + Compliance Table + Learner sections
Employee	Welcome + Stats + Continue Learning + Timeline + Achievements + Recommendations

3. New Plugins

Plugin	Purpose
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local_airpay_pages	Static pages: Privacy Policy, Terms of Use, Help Center, Contact Us
block_airpay_compliance	RAG compliance dashboard (Red/Amber/Green status per course)
local_airpay_lifecycle	Employee lifecycle: auto-enrol in mandatory courses on user creation
local_airpay_integrations	Integrations hub (placeholder for future HRMS/Slack/Teams)

4. BizLMS Bug Fixes (30+ files)

Category	Files Fixed	Issue
jQuery AMD	13 mustache templates	"\$ is not a function" — jQuery not available globally in Moodle 4.5
Typo fixes	9 lang files	"Bussiness Unit" → "Business Unit" (visible on every page)
Missing JS	1 JavaScript file	fulldescriptionpopover.js — blocked Online Exams + Classrooms pages
PHP fixes	2 PHP files	MESSAGE_DEFAULT_LOGGEDIN deprecated + use core_component warning
Permissions	2 DB records	Catalog viewcatalog PROHIBIT → ALLOW for admin/manager roles
Config	2 DB records	perfdebug 7→0, enableglobalsearch 1

Moodle Version Upgrade Impact

The production database was built on Moodle 4.1. Our local development runs Moodle 4.5.10. When the production database is used with 4.5 code, Moodle automatically upgrades its internal tables.

Plugin Upgrades (53 standard plugins)

All standard Moodle plugins upgrade from 2022112800 (4.1) to 2024100700 (4.5). This includes:

- All 24 activity modules (Assignment, Forum, Quiz, SCORM, etc.)
- All question types and behaviors
- All authentication and enrolment plugins
- All editors (Atto, TinyMCE)
- All report plugins
- All repository plugins

New Plugins Installed (30)

Moodle 4.5 introduces new standard plugins:

- Subsection activity module
- Ordering question type
- Code highlighter filter
- Multi-factor authentication (MFA) tool with 13 factor types
- AI placement plugins (course assist, editor) with Azure AI and OpenAI providers
- Communication providers (Custom link, Matrix)
- H5P framework v1.27

- Theme usage report
- Custom field number type
- HTML and Link TinyMCE plugins
- SMS gateway (AWS)

Legacy Plugins Deleted (21)

Removed deprecated plugins:

- Old assignment module (replaced by mod_assign)
- TinyMCE legacy editor + 9 plugins (replaced by new TinyMCE)
- Legacy log store
- memcached and mongodb cache stores
- H5P framework v1.24 (replaced by v1.27)
- tidy filter

One Fix Applied During Upgrade

- `local_airpay_lifecycle/db/messages.php` : Changed `MESSAGE_DEFAULT_LOGGEDIN` to `MESSAGE_DEFAULT_ENABLED` (constant removed in Moodle 4.5)

What Does NOT Change

Item	Status
Database schema	No changes needed — BizLMS tables and data preserved
User accounts	All 2,871+ users preserved with passwords, roles, enrolments
Course content	All 411 courses preserved with SCORM packages
Completion records	All 56,000+ completion records preserved
Certificates	All 8,822 issued certificates preserved
Costcenter structure	All 213 costcenters and 3 tenants preserved
Learning paths	All 20+ learning plans preserved
moodledata	Not touched — all user uploads, SCORM files, cache preserved

Risk Assessment

Risk	Likelihood	Impact	Mitigation
Theme rendering issue on production	Low	Medium	Tested with real production data (2,871 users). Quick rollback: change theme back to epsilon via SQL
BizLMS jQuery fix breaks existing feature	Low	Medium	All 13 templates tested. Fix is additive (wraps in require) — doesn't change logic
Plugin install fails on production	Low	High	Tested upgrade path from 4.1→4.5. One fix already applied (MESSAGE_DEFAULT_ENABLED)

User disruption during deploy	Medium	Low	Deploy during off-hours. Enable maintenance mode. Takes <30 minutes
SMTP not configured	High	Medium	Theme works without email. Configure SMTP separately

Production Data Tested

This release was tested against the actual production database (6th April 2026 backup):

Metric	Value	Verified
Active users	2,871	Login + dashboard for all 7 role types
Courses	411	Catalog + course cards + detail pages
Tenants	3 (Airpay, Public, ZEEA)	Tenant isolation + dynamic branding
Enrolments	39,000+	Employee stats + manager team data
Completions	8,000+	Certificate display + completion rates
Costcenters	213	Manage Company page + stat cards

Test Users Verified

Role	User	Tenant	Result
Siteadmin	academy@airpay.co.in	All	9/9 pages pass
L&D Admin	nitin.rajput@airpay.co.in	Airpay	Tenant-scoped dashboard
Manager	binay.upadhyay@airpay.co.in	Vyaapaar	Team KPIs + 9 direct reports
Employee	mithu.bala@airpay.co.in	Vyaapaar	48 enrolled, 21 completed, 15 certs
External	demoairpayacademy@gmail.com	Public	42 enrolled, tenant isolated
ZEEA	user.4156200@gmail.com	ZEEA	Dynamic logo + Swahili courses
Guest	(not logged in)	N/A	Homepage + login + registration

Deliverables Included in This Release

Document	Purpose
Install Guide (this package)	Step-by-step installation and upgrade instructions
Upgrade Summary (this document)	What changed and why
Deployment Runbook	21-point post-deploy verification checklist
Validation Matrix	144-item test coverage report
Production DB Diagnostic	33 questions for data health assessment

Stabilization Guide	74 duplicate courses + admin cleanup playbook
10x Roadmap	Future development plan (Tier 1-3)
Data Verification Excel	Orphaned users + never-logged-in + full roster
Upgrade Log Excel	105 plugin upgrade/install/delete records

Approval Required

This release is ready for production deployment pending management approval.

Recommended deployment approach:

1. Deploy to staging server first (H2)
2. Run 21-point verification checklist on staging
3. Schedule production deployment during off-hours (e.g., Saturday 2 AM IST)
4. Enable maintenance mode → deploy → verify → disable maintenance mode
5. Monitor for 48 hours post-deploy

Estimated downtime: 30-45 minutes (maintenance mode)
